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Abstract

Refraction causes random dynamic distortions in atmospheric turbulence and in views across a water interface. The later scenario is experienced by submerged animals seeking to detect prey or avoid predators, which may be airborne or on land. Man encounters this when surveying a scene by a submarine or divers, while wishing to avoid the use of an attention-drawing periscope. The problem of inverting random refracted dynamic distortions is difficult, particularly when some of the objects in the field of view are moving. On the other hand, in many cases, just those moving objects are of interest, as they reveal animal, human or machine activity. Furthermore, detecting and tracking these objects does not necessitate handling the difficult task of complete recovery of the scene. We show that moving objects can be detected very simply, with low false-positive rates, even when the distortions are very strong and dominate the object motion. Moreover, the moving object can be detected even if it has zero mean motion. While the object and distortion motions are random and unknown, they are mutually independent. This is expressed by a simple motion feature which enables discrimination of moving object points vs. the background.

Index Terms

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Independent Components in Dynamic Refraction



Fig. 1. A frame of an airborne scene, taken by a submerged camcorder. The underwater scene is typically darker and smoother. It undergoes total internal reflection (TIR) by the water-air interface (WAI) and appears in the lower part of the image. TIR is bounded by Snell's window, where the airborne scene is seen. The strong random distortion changes in space and time. The task is to be able to sense objects that move, and ignore static objects whose apparent motion is only due to dynamics of the medium.

1 Introduction

Changes in refractive index of a medium along a line of sight (LOS) cause distortion [21], [31], [41]. Such changes are created by atmospheric turbulence. They are also created when looking through a water-air interface (WAI). In a wavy WAI, the refracted view changes spatiotemporally, as in a turbulent atmosphere but more strongly, even in mild weather conditions. In either case, the result is random dynamic distortions. The distortions are particularly strong when a submerged viewer looks obliquely upwards through a WAI (Figs. 1,2). Due to Snell's Law, small changes in the WAI slope are amplified to large angular changes of the airborne LOS, particularly near the horizon.

The paper deals with vision under natural random distortions. We focus on shallow oblique views upwards through a wavy WAI, for several reasons. First, it is most challenging, since the distortions in this scenario are strongest. Second, it is related to biological vision: it is experienced by submerged animals seeking to detect prey or avoid predators, which are airborne or on land. The archer fish [40] shoots down airborne prey using a water jet, aiming while being submerged. It has also been hypothesized that seals watch for ambushing polar bears, prior to surfacing for air. There are other animals, such as crocodiles, that prey (or wish to avoid predators) across the WAI. There are also airborne and land animals [24], that detect and hunt fish by looking down through the WAI. In water, oblique upward vision through a WAI can function as a virtual periscope for submariners and divers, who wish to avoid using attention-drawing physical periscopes. In air, a possible application is long distance surveillance through atmospheric turbulence.

Computer vision in random distortions has been studied extensively, as we detail in Sec. 2. Most studies focused on inverting the refractions. This is difficult, particularly when some of the objects are moving. On the other hand, in many cases, those moving objects are the actual interest, as they reveal animal, human or machine activity. However, detecting and tracking these objects does not necessitate complete inversion of the medium and recovery of the scene. Deciding whether motion of an image feature is solely due to refractive distortions or a due to real object motion (compounded with distortion) is a *classification* problem. Given a good feature vector, classification can be obtained simply and computationally cheaply. This is the task of this paper.[1]

Discriminating consistently moving objects from a randomly moving background was studied in open air background subtraction [5], [26], [32], tracking [8], [47] and linear motion saliency [46]. In refractive distortions, however, moving objects are not *beside* a random background. Instead, the image of the moving object is prone to the same strong random distortions as static objects. In fact, motion due to distortions often dominates over object motion. In other words, the motion noise can be larger than the motion signal. As our numerical analysis and experiments show, inter-frame motion can span a significant portion of the field of view (FOV). This is much larger than work typically done in atmospheric turbulence [19], where local optical flow is used.

While the object and distortion motions are random and unknown, they are *mutually independent.* We exploit this fact. Being independent components, the location covariance due to object motion is *additive* to covariance stemming from medium dynamics. Hence, simple location covariance is a powerful classification feature. Characterizing the features corresponding to the class of static objects benefits from physics-based analysis of distortion statistics.

2 Related Work: Vision via a WAI

Approaches to vision through a WAI are:

1. Analysis of flat water distortion
2. Physics-based recovery of a wavy WAI

1. Trajectories are classified as relating to diffuse or specular reflection in [35].

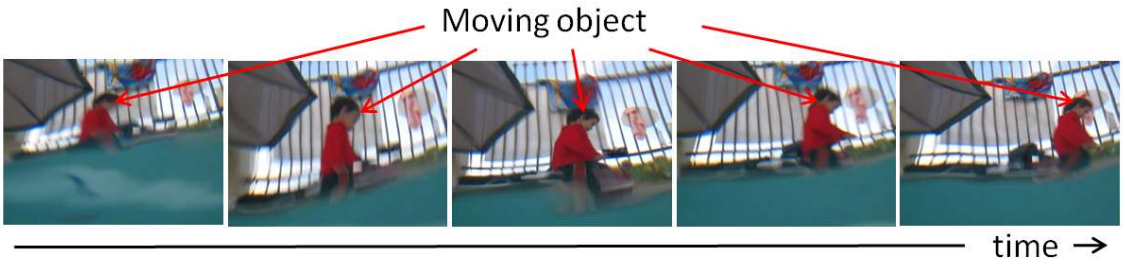


Fig. 2. Sample frames from a video. The entire observed scene is dynamically distorted. All objects undergo strong nonlinear and nonuniform stretching, shifting and curving. In addition, there are objects which have inherent motion, superimposed on the distortion.

3. Statistical recovery of static objects via a wavy WAI

1. Distortions in oblique view and stereo vision through a flat WAI were thoroughly analyzed [20], [36], [44]. Ref. [42] introduces optical elements to the camera, to correct for flat WAI distortion. The conclusion we have is that if a flat WAI is the only distortion source, then this distortion can be compensated for using stereo. Therefore, the remaining visual challenge is to overcome perturbations to the flat-WAI case. These are caused by random WAI waves.

2. To compensate for distortion caused by WAI waves, some methods suggest detection of local flat WAI instances, or dynamic reconstruction of the WAI. The former requires accumulation of sparse data from thousands of frames and a pulsed laser [27] synchronized to the camera. This complex and slow approach cannot handle moving objects. To recover the water geometry in real time, a theoretical work by Dolin [15] hypothesizes that the WAI slope can be inferred by its specular reflection[2] of a cloudless sky. Multi-cameras methods have also been applied to WAI recovery [14], [33], [37], [38], [45]. A critical review of passive approaches to recover the WAI[3] is in Ref. [22]. Those studies focused on viewing downwards into water. Looking up through the WAI to create a *virtual periscope* is theoretically proposed [30] based on a wave model constrained by self occlusions, detected by dark spots. Polarization-based WAI recovery for visual correction was also theoretically proposed [39].

3. Some methods bypass physical measurement of the WAI, and directly aim to recover static objects, using video. For small distortions, Ref. [34] uses optical flow to estimate the WAI normals, leading to image correction. There are patch-based versions of lucky imaging [17], [18]. They accumulate a large number of frames, out which a best-representative frame is selected per patch. Another patch-based approach seeks an optimal low dimensional parametric distortion model [43] to fit spatiotemporal data acquired with a fast camera. These approaches assume that the object is static while data is acquired. In our work, however, moving objects may exist in the FOV, and are actually a focus of interest.

2. Ref. [1] exploits the relative motion between a curved specular object and its environment to reconstruct its shape. This method is related to the recovery of specular surfaces from reflected distortions.

3. There are additional methods for measuring the water surface using reflection of structured light [3], [13]. They have not been tested as a tool for visual correction.

3 Theoretical Background

3.1 Snell Law in Vector Form

Let the LOS in water be along unit vector $\hat{\mathbf{v}}_{\mathrm{w}}$. It forms an angle θ_{w} with the WAI normal $\hat{\mathbf{N}}$, as illustrated in Fig. 3. At the WAI, the LOS refracts and proceeds in air along unit vector $\hat{\mathbf{v}}_{\mathrm{a}}$, which forms an angle θ_{a} with $\hat{\mathbf{N}}$. The vector form of Snell's refraction law [25] is

$$\hat{\mathbf{v}}_{\mathrm{a}} \times \hat{\mathbf{N}} = n\hat{\mathbf{v}}_{\mathrm{w}} \times \hat{\mathbf{N}}. \tag{1}$$

Here $\times$ is the cross product and $n \approx 1.33$ is the refraction index of water. Eq. (1) expresses the coplanarity of $\hat{\mathbf{v}}_{\mathrm{w}}$, $\hat{\mathbf{v}}_{\mathrm{a}}$ and $\hat{\mathbf{N}}$, and the familiar scalar Snell law

$$\sin\theta_{\mathrm{a}} = n \sin\theta_{\mathrm{w}} \ . \tag{2}$$

An additional useful vector form of the law [25] is

$$\hat{\mathbf{v}}_{\mathrm{a}} = n\hat{\mathbf{v}}_{\mathrm{w}} + \hat{\mathbf{N}}(\cos\theta_{\mathrm{a}} - n\cos\theta_{\mathrm{w}}), \tag{3}$$

where

$$\cos\theta_{\mathrm{w}} = \hat{\mathbf{v}}_{\mathrm{w}} \cdot \hat{\mathbf{N}}. \tag{4}$$

If $\theta_{\mathrm{w}} > \arcsin(1/n)$, no airborne object is projected along the LOS. Instead, total internal reflection (TIR) of the underwater scene occurs.

3.2 Surface Slope Statistics

Cox and Munk [12] comprehensively studied the WAI wave slope statistics. This is the basis of much of the work done in relation to reflection by the sea surface. Accordingly, the WAI gradient is approximately Gaussian. Let the height of the WAI be z_{s} in the laboratory reference frame. The laboratory coordinate system is (ξ, η, z). Specifically, $\hat{\mathbf{z}} = (0, 0, 1)$ points to the zenith and (ξ, η) are lateral (horizontal) coordinates. The WAI gradient is $\nabla z_{\mathrm{s}} = [\partial_\xi z_{\mathrm{s}}, \partial_\eta z_{\mathrm{s}}]^{\mathrm{T}}$, where T denotes transposition. The probability density distribution (PDF) of the WAI gradient is

$$\mathrm{PDF}(\nabla z_{\mathrm{s}}) \sim \exp\left[-\frac{1}{2}(\nabla z_{\mathrm{s}})^{\mathrm{T}} \mathbf{\Sigma}_{\mathrm{grad}}^{-1} \nabla z_{\mathrm{s}}\right] \ , \tag{5}$$

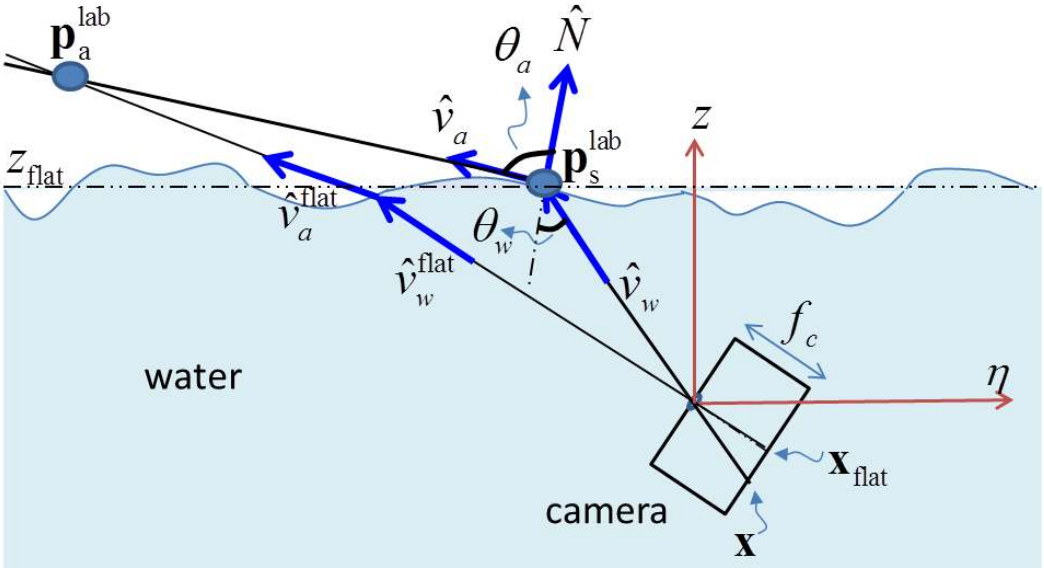


Fig. 3. Refraction at a wavy water surface.

where $\mathbf{\Sigma}_{\mathrm{grad}}$ is the covariance matrix of the WAI gradient. In non-directional wind, the matrix is $\mathbf{\Sigma}_{\mathrm{grad}} = \sigma_{\mathrm{grad}}^2 \mathbf{I}$, where $\mathbf{I}$ is the identity matrix and

$$\sigma_{\mathrm{grad}}^2 = 0.003 + 0.0051W, \tag{6}$$

Here W is the wind speed in $[\mathrm{meters/sec}]$.

Following the perturbation distribution of the medium as shown in Eq. (5), the surface normal $\hat{\mathbf{N}}$ and the medium distortion are random. This is discussed later in Sec. 5.1.

4 Distortion Model

We derive a distorted projection model. This model is the basis for defining theoretically the statistical class of static objects. If the WAI interface is completely flat, then the image is considered undistorted. We are only interested here in perturbations to that state. Consider an airborne object point at location $\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}}$ in the lab coordinate system (see Fig. 3). The system's origin is at the center of projection of a submerged camera. When the WAI is flat, $\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}}$ is projected to pixel $\mathbf{x}_{\mathrm{flat}}$ in the detector plane. When the WAI is wavy, $\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}}$ is projected to pixel $\mathbf{x}$. The distortion thus takes the form of a shift

$$\mathbf{s}^{\mathrm{medium}} = \mathbf{x} - \mathbf{x}_{\mathrm{flat}} \tag{7}$$

in the image coordinates corresponding to the object point.

We now explain how $\mathbf{s}^{\mathrm{medium}}$ is derived. The distorted projection $\mathbf{x}$ is the output sought in this section, where the flat water pixel $\mathbf{x}_{\mathrm{flat}}$ is the input. Given a pixel location $\mathbf{x}_{\mathrm{flat}}$ in the camera detector plane, Sec. 4.1 associates it with an airborne object point, $\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}}$. Once $\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}}$ is obtained, we can derive the distorted projection $\mathbf{x}$. In waves, the WAI slope varies, changing the refraction angle. When observing distant objects, this is the dominant distortion source. Moreover, the water height above the camera changes. In mild weather, the wave amplitude is orders of magnitude smaller than the distance to the objects, and has just a small consequence on distortion. Now, as illustrated in Fig. 3, a ray coming from $\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}}$ hits the WAI at an unknown surface point $\mathbf{p}_{\mathrm{s}}^{\mathrm{lab}}$ before entering the camera pupil. Thus, the refraction causes the object point $\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}}$ to be projected to the distorted pixel location $\mathbf{x}$. In Sec. 4.2 we derive the distorted projection $\mathbf{x}$, based on $\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}}$.

Hence, given a flat pixel $\mathbf{x}_{\mathrm{flat}}$, the distorted pixel $\mathbf{x}$ is calculated based on Sec. 4.1 and Sec. 4.2. Hence, $\mathbf{s}^{\mathrm{medium}}$ is given by Eq. (7). The derivation of $\mathbf{s}^{\mathrm{medium}}$ is used further in Sec. 5 to study the statistics of the medium distortions.

4.1 Derivation of Flat Back-Projection

This section explains the calculation of the airborne object point $\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}}$ (Fig. 3) that is projected to a given pixel $\mathbf{x}_{\mathrm{flat}}$, when the WAI is calm. The obtained $\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}}$ is used later in Sec. 4.2 to derive its distorted projection $\mathbf{x}$ when the WAI is wavy. Consider Fig. 3. In the 3D coordinate system of the camera, the origin is at the center of projection. Pixel $\mathbf{x}_{\mathrm{flat}}$ on the detector is at location

$$\mathbf{p}_{\mathbf{x}_{\mathrm{flat}}}^{\mathrm{cam}} = \begin{bmatrix} \mathbf{x}_{\mathrm{flat}} \\ f_{\mathrm{c}} \end{bmatrix}, \tag{8}$$

where f_{c} is the focal length of the camera in units of pixels.[4]

In the lab coordinate system, the 3D location of the detector pixel is

$$\mathbf{p}_{\mathbf{x}_{\mathrm{flat}}}^{\mathrm{lab}} = \mathbf{R}^{-1}(\mathbf{p}_{\mathbf{x}_{\mathrm{flat}}}^{\mathrm{cam}}). \tag{9}$$

Here $\mathbf{R}$ is the camera rotation matrix (it encapsulates the extrinsic parameters of the camera, since the two coordinate

4. The sign of f_{c} follows the common notation in which the detector plane virtually lies in front of the camera pupil.

systems share the origin). Thus, in the lab coordinate system, the submerged LOS is along unit vector

$$\hat{\mathbf{v}}_{\mathrm{w}}^{\mathrm{flat}} = \mathbf{p}_{\mathbf{x}_{\mathrm{flat}}}^{\mathrm{lab}} / \| \mathbf{p}_{\mathbf{x}_{\mathrm{flat}}}^{\mathrm{lab}} \| . \tag{10}$$

Based on Eq. (2), $\sin^2\theta_{\mathrm{a}} = n^2 \sin^2\theta_{\mathrm{w}}$, thus

$$\cos\theta_{\mathrm{a}} = \sqrt{1 - n^2 + n^2 \cos^2\theta_{\mathrm{w}}} . \tag{11}$$

Substituting Eqs. (4) and (11) into Eq. (3) and setting $\hat{\mathbf{N}} = \hat{\mathbf{z}}$ yields the airborne LOS direction

$$\hat{\mathbf{v}}_{\mathrm{a}}^{\mathrm{flat}} = n \hat{\mathbf{v}}_{\mathrm{w}}^{\mathrm{flat}} + \hat{\mathbf{z}} \left[\sqrt{1 - n^2 + n^2 (\hat{\mathbf{v}}_{\mathrm{w}}^{\mathrm{flat}} \cdot \hat{\mathbf{z}})^2} - n \hat{\mathbf{v}}_{\mathrm{w}} \cdot \hat{\mathbf{z}} \right] . \tag{12}$$

Now we can set the above-surface object point $\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}} = (x_{\mathrm{a}}, y_{\mathrm{a}}, z_{\mathrm{a}})^{\mathrm{T}}$. Back-propagating from the camera's center of projection, the refracted LOS is a sum of a submerged LOS segment and an airborne one. Thus,

$$\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}} = \hat{\mathbf{v}}_{\mathrm{w}}^{\mathrm{flat}} l_{\mathrm{w}}^{\mathrm{flat}} + \hat{\mathbf{v}}_{\mathrm{a}}^{\mathrm{flat}} l_{\mathrm{a}}^{\mathrm{flat}} , \tag{13}$$

where $l_{\mathrm{w}}^{\mathrm{flat}}$ and $l_{\mathrm{a}}^{\mathrm{flat}}$ are line parameters, which we now derive. Let the camera be at depth z_{flat} below the flat WAI and the object at height z_{a} above the flat WAI. Then,

$$l_{\mathrm{w}}^{\mathrm{flat}} = z_{\mathrm{flat}} / v_{\mathrm{w},z}^{\mathrm{flat}} \quad , \quad l_{\mathrm{a}}^{\mathrm{flat}} = z_{\mathrm{a}} / v_{\mathrm{a},z}^{\mathrm{flat}} . \tag{14}$$

Here $v_{\mathrm{w},z}^{\mathrm{flat}}$ and $v_{\mathrm{a},z}^{\mathrm{flat}}$ are, respectively, the vertical components of $\hat{\mathbf{v}}_{\mathrm{w}}^{\mathrm{flat}}$ and $\hat{\mathbf{v}}_{\mathrm{a}}^{\mathrm{flat}}$.

4.2 Derivation of Distorted Projection

This section derives the distorted projection $\mathbf{x}$ of the airborne object point $\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}}$. The point $\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}}$ was derived in Sec. 4.1. The obtained distorted pixel $\mathbf{x}$ is used together with the given undistorted pixel $\mathbf{x}_{\mathrm{flat}}$ to calculate the medium distortion according to Eq. (7). When the WAI is wavy, a ray from $\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}}$ hits the WAI at location $\mathbf{p}_{\mathrm{s}}^{\mathrm{lab}}$, in the laboratory coordinate system, as illustrated in Fig. 3. The airborne LOS is along unit vector vector

$$\hat{\mathbf{v}}_{\mathrm{a}} = (\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}} - \mathbf{p}_{\mathrm{s}}^{\mathrm{lab}}) / \| \mathbf{p}_{\mathrm{a}}^{\mathrm{lab}} - \mathbf{p}_{\mathrm{s}}^{\mathrm{lab}} \| . \tag{15}$$

Let point $\mathbf{p}_{\mathrm{s}}^{\mathrm{lab}}$ be at height z_{s} above the camera. The lateral coordinates of $\mathbf{p}_{\mathrm{s}}^{\mathrm{lab}}$ are unknown. There is an additional constraint, however, to $\mathbf{p}_{\mathrm{s}}^{\mathrm{lab}}$: it is on a submerged LOS of the camera:

$$\mathbf{p}_{\mathrm{s}}^{\mathrm{lab}} = \hat{\mathbf{v}}_{\mathrm{w}} l_{\mathrm{w}} . \tag{16}$$

Here

$$l_{\mathrm{w}} = z_{\mathrm{s}} / v_{\mathrm{w},z} \tag{17}$$

where $v_{\mathrm{w},z}$ is the vertical component of $\hat{\mathbf{v}}_{\mathrm{w}}$. Using Eqs. (16,17) in (15),

$$\hat{\mathbf{v}}_{\mathrm{a}} = \frac{\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}} - \hat{\mathbf{v}}_{\mathrm{w}} z_{\mathrm{s}} / v_{\mathrm{w},z}}{\| \mathbf{p}_{\mathrm{a}}^{\mathrm{lab}} - \hat{\mathbf{v}}_{\mathrm{w}} z_{\mathrm{s}} / v_{\mathrm{w},z} \|} . \tag{18}$$

Note that $\hat{\mathbf{v}}_{\mathrm{a}}$ and $\hat{\mathbf{v}}_{\mathrm{w}}$ are unknown, but they are related by Snell's law, given the surface normal. Substituting Eqs. (4) and (11) into Eq. (3),

$$\hat{\mathbf{v}}_{\mathrm{a}} = n \hat{\mathbf{v}}_{\mathrm{w}} + \hat{\mathbf{N}} \left[\sqrt{1 - n^2 + n^2 (\hat{\mathbf{v}}_{\mathrm{w}} \cdot \hat{\mathbf{N}})^2} - n \hat{\mathbf{v}}_{\mathrm{w}} \cdot \hat{\mathbf{N}} \right] . \tag{19}$$

The two equations (18,19) provide two constraints for solving the degrees of freedom of the unit vector $\hat{\mathbf{v}}_{\mathrm{w}}$. We obtained the solution using Matlab's solver.

Once $\hat{\mathbf{v}}_{\mathrm{w}}$ is obtained, we can derive the distorted projection. The shifted detector location, onto which the object is projected is on the LOS, thus in direction $\hat{\mathbf{v}}_{\mathrm{w}}$ from the center of projection. In the camera coordinate system, this direction vector is $\hat{\mathbf{v}}_{\mathrm{w}}^{\mathrm{cam}} = \mathbf{R} \hat{\mathbf{v}}_{\mathrm{w}}$. Thus, in the camera coordinate system, the 3D location of the detector point is

$$\mathbf{p}_{\mathbf{x}}^{\mathrm{cam}} = \hat{\mathbf{v}}_{\mathrm{w}}^{\mathrm{cam}} l_c , \tag{20}$$

where l_c is a line parameter, which we now derive. The shifted pixel is $\mathbf{x}$. Similarly to (8), $\mathbf{x}$ corresponds to 3D position

$$\mathbf{p}_{\mathbf{x}}^{\mathrm{cam}} = \begin{bmatrix} \mathbf{x} \\ f_{\mathrm{c}} \end{bmatrix} . \tag{21}$$

Hence,

$$l_c = f_{\mathrm{c}} / v_{\mathrm{w},z}^{\mathrm{cam}} \tag{22}$$

to satisfy consistency of (20) with the third component in Eq. (21), while $v_{\mathrm{w},z}^{\mathrm{cam}}$ is the third component of $\hat{\mathbf{v}}_{\mathrm{w}}^{\mathrm{cam}}$. Eqs. (20,22) determine $\mathbf{p}_{\mathbf{x}}^{\mathrm{cam}}$. The distorted projection pixel $\mathbf{x}$ comprises the first two components of $\mathbf{p}_{\mathbf{x}}^{\mathrm{cam}}$, according to Eq. (21).

5 Stochastic Model

5.1 Distortion Statistics

Section 4 derives $\mathbf{s}^{\mathrm{medium}}$, for a given WAI normal $\hat{\mathbf{N}}$ and height z_s. The normal is related to the WAI gradient by

$$\hat{\mathbf{N}} = \begin{bmatrix} -\nabla z_{\mathrm{s}} \\ 1 \end{bmatrix} [(\nabla z_{\mathrm{s}})^2 + 1]^{-1/2} . \tag{23}$$

Due to the dynamics of the medium, ∇z_{s} is random, distributed according to Eq. (5). Hence, the consequent distortion $\mathbf{s}^{\mathrm{medium}} = (s_x^{\mathrm{medium}}, s_y^{\mathrm{medium}})$ is random, and its distribution can be drawn from the perturbation distribution of the medium (Sec. 3.2) and the distortion model (Sec. 4). Our experiments (see Sec. 7), were conducted in swimming pools. There, distortion statistics are determined by water depth, the size of the pool and the fact that the waves were created as a result of swimmers rather than wind. Nevertheless, in this general study of distortion statistics, the sea surface statistical model of Cox and Munk (Sec. 3.2) is taken as a representative. While this is an approximation to the experimental setup, it relies on a well established theory and, as will be seen in Sec. 7, yields good experimental results in our task.

Consider a typical setup. Let a camera have a focal length of 6.5 mm. The camera is submerged at a depth of $z_{\mathrm{flat}} = 20$ cm below the WAI. Underwater, it stares upwards at 40° from the zenith, corresponding to 31° above the airborne horizon. Object point $\mathbf{p}_{\mathrm{a}}^{\mathrm{lab}}$ is at $z_a = 1.5$ meters above the WAI. WAI slopes are then drawn from the distribution (5). In addition, the WAI height z_{s} is uniformly distributed around $z_{\mathrm{flat}} \pm 2$ cm. We found, as expected, that the wave amplitude range has only a marginal effect on the image distortion statistics, which are rather dominated by

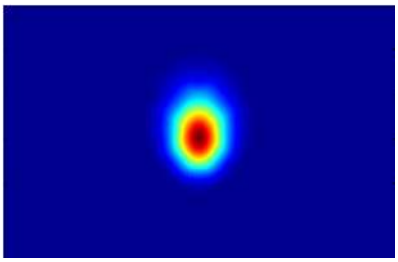


Fig. 4. Numerically calculated distribution of $\mathbf{x}$, for a submerged camera close to the WAI, looking obliquely upwards. WAI waves perturb $\mathbf{x}$. The waves are distributed according to Ref. [12], corresponding to a nondirectional 4 knot wind.

the WAI gradients. Let $\mathbf{x}_{\text{flat}}$ be at the center of the FOV. The x-axis of the camera is perpendicular to the z-axis of the laboratory coordinate system. Hence, there is symmetry around the y axis of the image. The simulated distribution of the distorted $\mathbf{x}$ is shown in Fig. 4.

Regardless of the distortion model, we can define the covariance matrix of $\mathbf{s}^{\text{medium}}$

$$\mathbf{\Sigma}^{\text{medium}} = \begin{bmatrix} (\sigma_x^{\text{medium}})^2 & \text{OD} \\ \text{OD} & (\sigma_y^{\text{medium}})^2 \end{bmatrix}, \tag{24}$$

where the off diagonal (OD) elements are expectation over time t of $[s_x^{\text{medium}}(t) s_y^{\text{medium}}(t)]$. Here σ_x^{medium} and σ_y^{medium} denote the respective standard deviations (STDs) in the respective image axes. In the example of Fig. 4, OD is null, $\sigma_x^{\text{medium}} \approx 130$ and $\sigma_y^{\text{medium}} \approx 170$ pixels.

5.2 Dependence on the Viewing Angle

Let θ_a^{flat} be the LOS angle in air, relative to the zenith, when the water surface is flat. This is also the average of the angle θ_a over time (see Fig. 3). Towards the airborne horizon, θ_a^{flat} increases. Let an airborne object be viewed near grazing angles to the horizon. Due to Snell's Law, even small perturbations in the slope of the WAI result in significant variance of $\mathbf{s}^{\text{medium}}$. On the other hand, when looking straight upwards (zenith), WAI waves yield only mild changes of the airborne LOS.

Sec. 5.1 presented the derivation of $\mathbf{\Sigma}^{\text{medium}}$, for a given point in the FOV and wave conditions (wind speed W). Indeed, σ_x^{medium} and σ_y^{medium} are functions of θ_a^{flat} and W. This dependency can be assessed by simulations in a range of different θ_a^{flat} and wind conditions. Our simulations show that as θ_a^{flat} increases, the pixel variance increases, particularly in the vertical pixel coordinate.

Wind Invariance

Our simulations show that similar dependence is observed for different values of W. Define normalized STDs as the ratios

$$r_x = \frac{\sigma_x^{\text{medium}}}{\sigma_x^{\text{medium}}\left(\theta_a^{\text{flat}}=0^o\right)}, \quad r_y = \frac{\sigma_y^{\text{medium}}}{\sigma_y^{\text{medium}}\left(\theta_a^{\text{flat}}=0^o\right)}. \tag{25}$$

Plotting r_x and r_y, reveals that their dependence on the W is weak. This observation allows r_x and r_y to be each well approximated by a representative function, independent of W. We thus simply averaged either r_x and r_y, for a range of W. The respective results $\bar{r}_x$ and $\bar{r}_y$ depend solely on θ_a^{flat}. Fig. 5 plots $\bar{r}_x$ and $\bar{r}_y$. Notice that $\bar{r}_x$ and $\bar{r}_y$ increase monotonically with θ_a^{flat}.

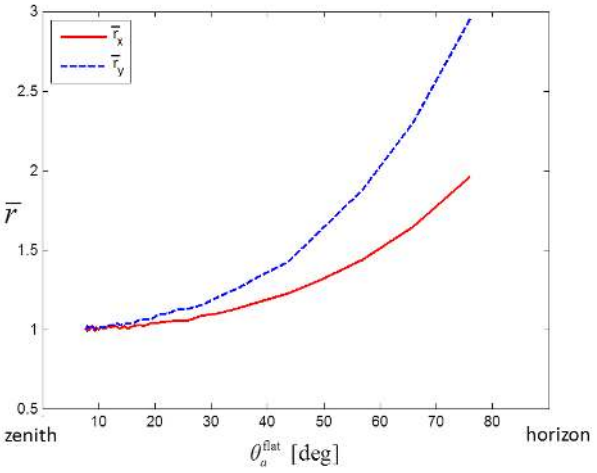


Fig. 5. Numerically calculated pixel STD ratios $\bar{r}_x$ and $\bar{r}_y$ as a function of angle θ_a^{flat} relative to zenith.

As we explain in the next sections, pixel shift variances are used by us as features for classification of static and dynamic objects. Thus, the dependence on the viewing angle has to be accounted and compensated for. Sec. 6 describes the feature calibration procedure by exploiting the theoretical $\bar{r}_x$ and $\bar{r}_y$ plotted in Fig. 5.

5.3 Observation Statistics: Independence

If the medium is calm, a moving object induces a shift of $\mathbf{s}^{\text{object}}$ pixels in its image coordinates. When both the object and the medium are dynamic, the projection of the object shifts by the vector

$$\mathbf{s}^{\text{observe}} = \mathbf{s}^{\text{medium}} + \mathbf{s}^{\text{object}}. \tag{26}$$

A static object is expressed by the special case $\mathbf{s}^{\text{object}} = 0$. Due to the random and fast kinetics of $\mathbf{s}^{\text{medium}}$, it is unlikely that the object would plan and perform upcoming moves in a way which linearly depends on forecasted medium kinetics. The shifts $\{\mathbf{s}^{\text{medium}}, \mathbf{s}^{\text{object}}\}$ are thus generally *independent* of each other. The independence of components in Eq. (26) is a key for our solution.

Due to the linear independence of the stochastic vectors $\mathbf{s}^{\text{medium}}$ and $\mathbf{s}^{\text{object}}$, the covariance matrix of the location of an observed object is

$$\mathbf{\Sigma}^{\text{observe}} = \mathbf{\Sigma}^{\text{medium}} + \mathbf{\Sigma}^{\text{object}}, \tag{27}$$

where $\mathbf{\Sigma}^{\text{object}}$ is the covariance matrix of $\mathbf{s}^{\text{object}}$. Hence, object motion increases the diagonal elements of $\mathbf{\Sigma}^{\text{observe}}$. Thus, the diagonal values of $\mathbf{\Sigma}^{\text{observe}}$ are simple *features* for classifying moving objects and stationary ones.

In practice, $\mathbf{\Sigma}^{\rm observe}$ is estimated based on the image data. Let $p = 1....N_{\rm points}$ index object points. The observed (distorted) position of each object point is denoted by $\mathbf{x}_p = [x_p, y_p]^{\rm T}$. It is tracked during $N_{\rm frames}$ frames. The mean position of each point is

$$\bar{\mathbf{x}}_p \equiv [\bar{x}_p, \bar{y}_p]^{\rm T} = \frac{1}{N_{\rm frames}} \sum_{t=1}^{N_{\rm frames}} \mathbf{x}_p(t) \ . \tag{28}$$

The estimated covariance matrix of each point is

$$\hat{\mathbf{\Sigma}}_p^{\rm observe} = \frac{1}{N_{\rm frames}-1} \sum_{t=1}^{N_{\rm frames}} [\mathbf{x}_p(t) - \bar{\mathbf{x}}_p][\mathbf{x}_p(t) - \bar{\mathbf{x}}_p]^{\rm T}. \tag{29}$$

The feature vector of point p is comprised of the diagonal terms in $\hat{\mathbf{\Sigma}}_p^{\rm observe}$, i.e., the estimated horizontal and vertical trajectory variances,

$$\mathbf{c}_p = [\hat{\sigma}^2_{{\rm x},p},\ \hat{\sigma}^2_{{\rm y},p}]^{\rm T} \ . \tag{30}$$

For simplicity, Eq. (30) ignores the OD terms of $\hat{\mathbf{\Sigma}}_p^{\rm observe}$. As shown in Sec. 5.1, the OD terms of $\mathbf{\Sigma}^{\rm medium}$ are null, at the camera optical axis. Due to projection on the detector plane, in off axis positions, the OD terms are non-zero and small. Accounting for the known camera projection, the location distribution can be corrected for this. Hence, the OD terms can be, theoretically, useful features: substantial OD values indicate a non-static object. However, this property can seldom be used: almost always, objects of interest move horizontally (walking, driving, sailing etc.) and sometimes they move vertically (animals at the water level drink from the water source, people load or unload gear, launch a spear in the water etc.). Either of these motions has very small OD covariance. Hence, the OD elements of moving and static objects are not expected to be significant, in most cases. Nevertheless, our subsequent analysis can be applied to feature vectors comprising the OD elements as well.

The elements of $\mathbf{c}_p$ are *not* the true variances of the trajectory, since they are not based on infinite $N_{\rm frames}$. They are *estimates*, based on a small $N_{\rm frames}$. As such, each $\mathbf{c}_p$ is a random sample from a distribution around the feature vector that would have been obtained had $N_{\rm frames}$ been infinite. The distribution width indicates the uncertainty of $\mathbf{c}_p$. Cochran's Theorem [9] states that if the observations are independent and normally distributed, their variance follows a scaled χ^2 distribution. Our analysis in Sec. 5 derives the distribution of $\mathbf{s}^{\rm medium}$: it is not exactly Gaussian, but it resembles it. Approximating the PDF of $\mathbf{s}^{\rm medium}$ as Gaussian, the distribution of the elements of $\mathbf{c}_p$ is $\sim \chi^2_{N_{\rm frames}-1}$, following Cochran's Theorem. Thus, the expectation and STD of the first element in Eq. (30), $c_p(1) = \hat{\sigma}^2_{{\rm x},p}$, are respectively

$$E[c_p(1)] = \sigma^2_{{\rm x},p} \ , \tag{31}$$

and

$$\mathrm{STD}[c_p(1)] = \frac{\sqrt{2}\sigma^2_{{\rm x},p}}{\sqrt{N_{\rm frames}-1}} \approx \frac{\sqrt{2}c_p(1)}{\sqrt{N_{\rm frames}-1}}, \tag{32}$$

with analogous expressions for $c_p(2) = \hat{\sigma}^2_{{\rm y},p}$.

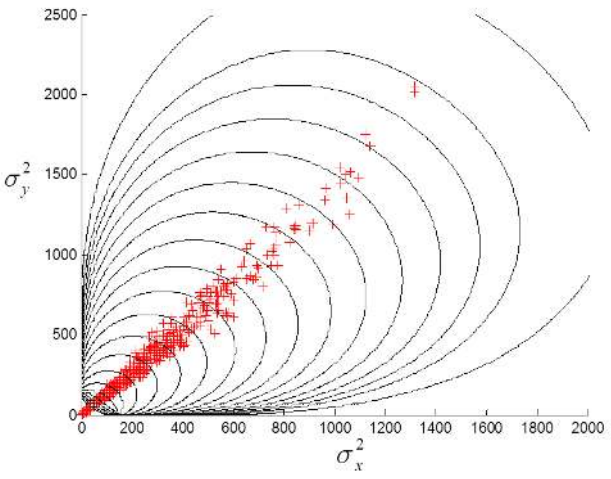


Fig. 6. Crosses mark a scatter plot of static-objects in feature-space, theoretically derived for wind conditions up to 2.5 knots. The plot is overlaid on a PDF. The contours mark level sets of the estimated PDF, and can be used as confidence intervals and thresholds.

5.4 Physics-based Feature Distribution

The distribution of static objects can be predicted, based on our theory. The inclination angle, focal length, FOV and depth of the camera are all fixed and known. A major unknown is the wind speed, which determines the WAI slope statistics. It also sets the WAI height distribution (which has just a marginal effect), according to the *Beaufort scale of wind force* [4].

Sec. 5.1 describes the derivation of $\mathbf{\Sigma}^{\rm medium}$, for a given point in the FOV and wind speed. Hence, spreading points $\mathbf{x}_{\rm flat}$ across the FOV, and sampling wind according to some distribution yields a theoretical scatter plot in feature space, corresponding to static objects. As an example, we simulated vectors $\mathbf{c}_p$ in a range of the unknowns. The static object points $\mathbf{x}_{\rm flat}$ correspond to views from 10° to 75° above the airborne horizon. The wind speed is sampled uniformly from 0 to 2.5 knots. The scatter plot is shown as red crosses in Fig. 6. The scatter plot is mainly concentrated in the lower-left part of the feature space. It extends to larger coordinates in feature space, sparsely. The extent depends on the maximum wind speed we allow.

6 Camera and Feature Calibration

To analyze experimental movies, we first need to estimate two camera parameters that affect the pixel variance: the underwater effective focal length, and the inclination angle of the camera. To compute the effective focal length of the camera in water, a submerged object of a known length $d^{\rm obj}$ is captured by the submerged camera, from a known distance $D^{\rm water}$. The image corresponding to the projected object is $d^{\rm pix}$ pixels long. Based on geometrical considerations, the focal length is

$$f_c = D^{\rm water} \frac{d^{\rm pix}}{d^{\rm obj}} \ , \tag{33}$$

in units of pixels.

The inclination angle of the camera is determined by the location of Snell's window in the FOV. This is detailed in Sec. 6.2. As a result of knowing both the inclination and f_c, it is now possible to determine θ_a^{flat}, for each camera pixel. This too, is derived in Sec. 6.2.

6.1 Feature Scaling

Our analysis in Sec. 5.2 shows that σ_x^{medium} and σ_y^{medium} inflate with increase of θ_a^{flat}. This inflation is known, and is given the multiplicative factors $\bar{r}_x$ and $\bar{r}_y$, plotted in Fig. 5. Furthermore, any image is an optical magnification which scales linearly with f_c, and so do σ_x^{medium} and σ_y^{medium}. We seek to use a single classifier for all moving pixels, in all movies. Therefore, all features $\mathbf{c}_p$ (both during training and testing) need to be scaled, to compensate for these systematic factors. Since generally $\bar{r}_x \neq \bar{r}_y$, scaling is different in each axis. Thus, the normalized features are

$$\widetilde{\sigma}^2_{\text{x},p} = \frac{\hat{\sigma}^2_{\text{x},p}}{[\bar{r}_x(p) f_c]^2} \ , \quad \widetilde{\sigma}^2_{\text{y},p} = \frac{\hat{\sigma}^2_{\text{y},p}}{[\bar{r}_y(p) f_c]^2} \ . \tag{34}$$

The factors $\bar{r}_x(p)$ and $\bar{r}_y(p)$ are individual to each feature p, as we explain. Eq. (28) estimates the mean vertical pixel location $\bar{y}_p$. This approximates the vertical location of the tracked image feature, had the water been flat and the object static. Sec. 6.2 derives the relation between the vertical pixel location $\bar{y}_p$ in flat water, and $\theta_a^{\text{flat}}(p)$. Then, Fig. 5 yields $\bar{r}_x(p)$ and $\bar{r}_y(p)$.

6.2 From image pixel to θ_a^{flat}

This section derives the airborne angle θ_a^{flat}, as a function of the vertical pixel coordinate. This is required for feature scaling, as explained in Sec. 6.1. By using the precalibrated camera focal length f_c, the total FOV angle of the camera is

$$\alpha^{\text{FOV}} = 2 \arctan\left(\frac{H}{2 f_c}\right) \ , \tag{35}$$

where H is the image height in pixels. All the related variables are presented in Fig. 7.

We can infer the average vertical location $\bar{y}^{\text{horizon}}$ of the airborne horizon, by temporally averaging the video (See Fig. 8). The airborne horizon corresponds to Snell's window underwater, i.e., the critical angle $\theta_c = 48.7^o$ in the lab coordinate system. All trajectories for which the mean vertical coordinate lies below $\bar{y}^{\text{horizon}}$ belong to the TIR region. Let h be the vertical extent of the TIR region (see Figs. 7 and 8) in the image. In the camera coordinate system, the view angle β_c corresponding to the airborne horizon is a fraction of the total FOV angle α^{FOV}

$$\beta_c \approx \alpha^{\text{FOV}} \frac{h}{H} \ . \tag{36}$$

Let object feature p be tracked in the video. Its mean vertical image coordinate is $\bar{y}_p$. Similarly to Eq. (36), this coordinate is related to view angle

$$\alpha_p \approx \alpha^{\text{FOV}} \frac{H - \bar{y}_p}{H} \tag{37}$$

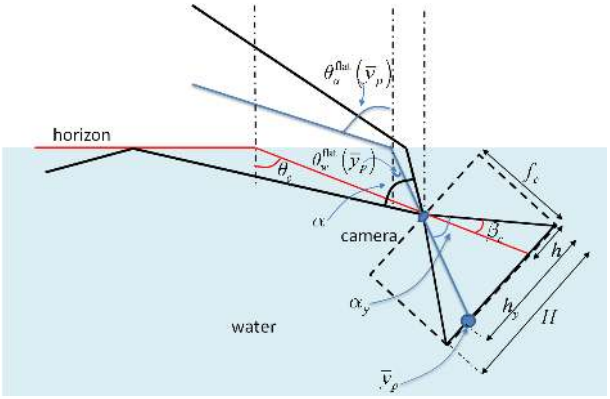


Fig. 7. Refraction at water surface.

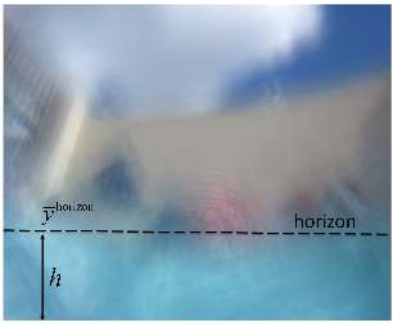


Fig. 8. Mean image of a video (best viewed in color). The horizon line is marked.

in the camera coordinate system. This is translated to the incident angle in underwater lab coordinate system,

$$\theta_w^{\text{flat}}(p) = \theta_c - (\alpha_p - \beta_c) \ . \tag{38}$$

Finally, using Snell's law

$$\theta_a^{\text{flat}}(p) = \arcsin[n \sin \theta_w^{\text{flat}}(p)] \ . \tag{39}$$

7 Classification Experiments

To show that $\mathbf{c}_p$ is a good feature vector for classifying static objects and detecting moving ones, we conducted experiments in different swimming pools, using different video-recording cameras (Canon PowerShot D10 mounted on a tripod, and unmounted Canon HV-30), at various focal length settings. Swimming pools have uncontrolled waves created by swimmers, and the wave statistics change in time as swimmers pass near or far. We attach some movies in the supplemental material.

Each movie is several seconds long, but we seek demonstrating fast response and dynamic detection of moving objects. So, we used $N_{\text{frames}} \approx 30 - 40$ in a temporally-sliding window, to estimate the trajectory statistics each time. In other words, about a second of acquisition was allocated to the task of detecting moving objects during

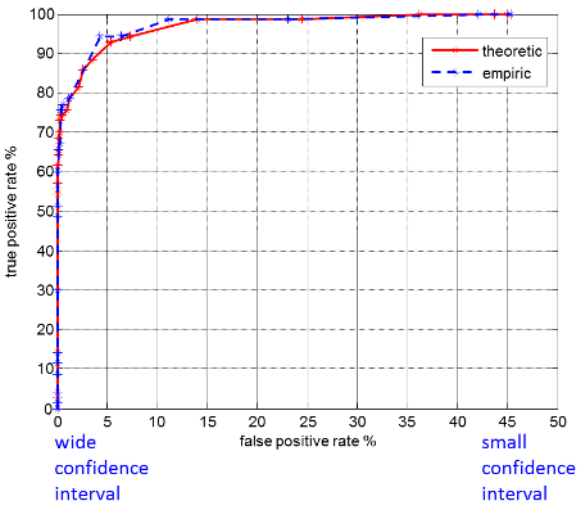


Fig. 9. ROC curve for different confidence interval threshold levels. A low false positive rate corresponds to wide static confidence interval, while a high false positive rate corresponds to a small interval.

dynamic refraction. Interest points were tracked, as described in Sec. 8. Figure 10 shows examples extracted from real movies. Each feature in the scatter plot is plotted as an ellipse: its center is at $\mathbf{c}_p$, while its widths are set according to Eq. (32). Red ellipses represent static object points while green ellipses represent dynamic object points. The corresponding red and green points are superimposed on the sample movie frames. Clearly, in these examples and others, moving objects are clearly separated from the cluster that encapsulates static objects in this feature space. Hence, classification in this feature space should usually not be difficult.

For classification, a decision contour in this feature space should be determined. Towards this, the feature distribution of static objects needs to be characterized.

7.1 Theoretical Classification

The distribution of static objects can be assessed from the theoretical simulation as was explained in Sec. 5.4. Overall, the estimated joint PDF in feature space is illustrated in Fig. 6, using its level sets. Each level in Fig. 6 is also a confidence interval: a certain percentage of static objects are expected to fall within the interval. Each of these contours can be selected as the classifier: the wider it is, the smaller is the expected number of false-positives (static objects falsely classified as moving), in testing. On the other hand, the wider the confidence interval, the larger is the expected number of false-negatives (moving objects falsely classified as static). This can be demonstrated by an ROC [48] curve. Fig. 9 shows an ROC curve (solid red line) derived from the theoretical PDF of static object features.

In our tests (e.g., Fig. 10), the classifier was based on thresholding on a confidence interval of the theoretical PDF.

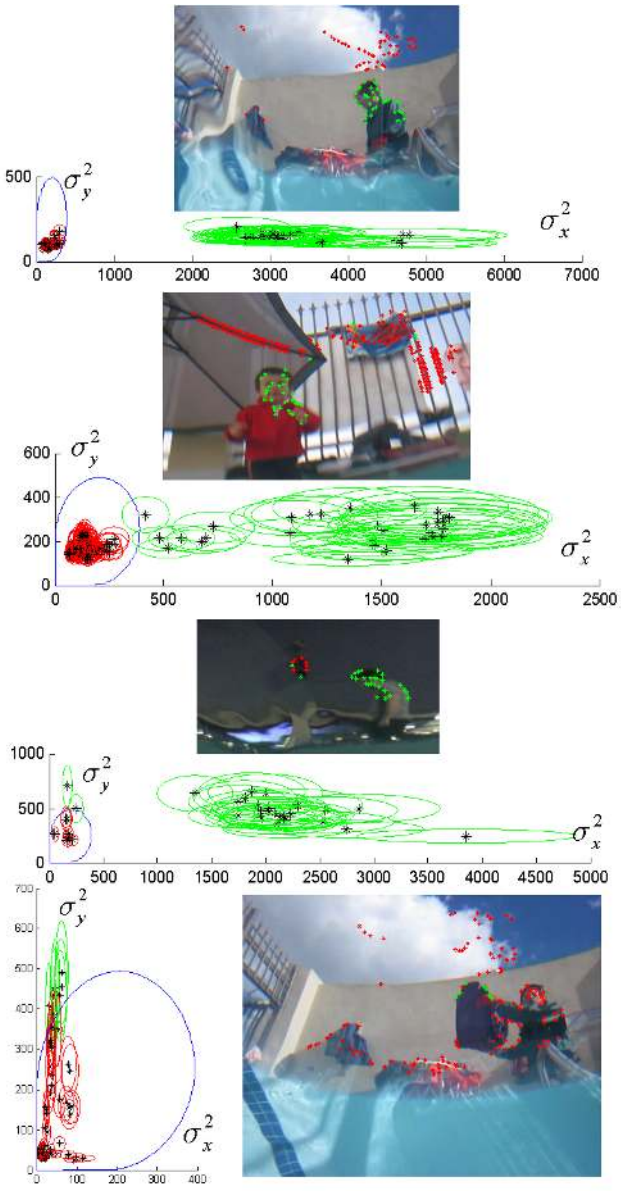


Fig. 10. The fixed theoretical classifier contour in feature space, and scatter plots of several test-sets. [Upper] Horizontally walking subject. [Lower] Vertically moving object.

The threshold corresponds to ∼85% true positive rate in Fig. 9. The classifying curve is shown as a blue line in Figure 10. Due to the fixed classifier setting, some examples yield some false positives or negatives. Tuning the classifier threshold eliminates many of these errors.

7.2 Empirical Training

The distribution of static objects can also be assessed by empirical training. We used some of the acquired movies to track static points. Objects were at various lines of sight (locations in the FOV), elevation angles and backgrounds. Different wave statistics occurred, as mentioned above. Overall, across the different movies, a total of 1000 scene points were used to characterize the static-object class. Each

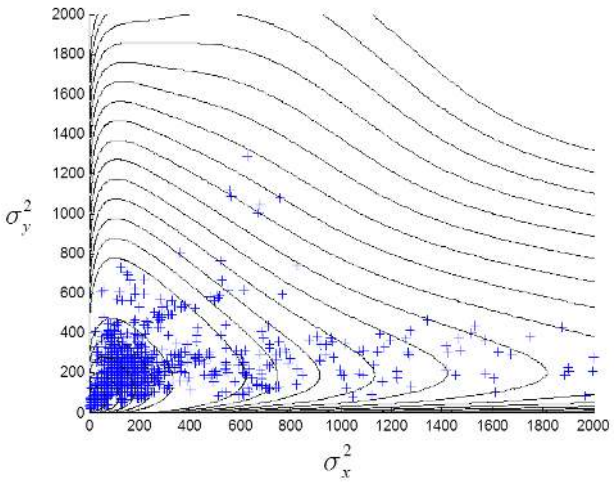


Fig. 11. Crosses mark a scatter plot of static-objects in feature-space, empirically derived from a set of movies in a pool. The plot is overlaid on a PDF. The contours mark level sets of the estimated PDF, and can be used as confidence intervals and classification thresholds.

measured $\mathbf{c}_p$ in the training set contributed a probability distribution to the feature space, as explained in Sec. 5.3. Similarly to the theoretical classifier, each of the contours in Fig. 11 can be selected as a classifier. Fig. 9 plots (dashed blue line) the ROC curve of the empirically trained classifier.

In the testing examples, we used the *same* empirical classifier ($\sim$80% true positive rate in Fig. 9), i.e., the same contour in feature space in all the tests. It is shown as a blue line in Figure 12.

Classification failure can occur for dynamic objects that have very small movements during the acquisition time. Because of the small movements, such an object may be classified as static because its feature-vectors fall inside the static-object threshold curve. Since the feature-vectors of such moving objects resemble the feature-vectors of water fluctuations, they cannot be separated by the variance features alone. Thus, additional features should be used. Classification failure may also occur as a result of tracking errors as we explain at the end of Sec. 8.

8 Tracking

The empirical process relies on tracking. *Interest points* (corners) corresponding to object points need to be tracked. The tracking needs to work despite severe disturbances, one of which is strong dynamic distortions, which we modeled in the sections above. Another disturbance is occasional loss of sight of the object point (regardless of being static or moving). This is caused by several phenomena. First, dynamic refraction sometimes causes strong image blur, when the refraction changes very rapidly in space and time, leading to smear within the exposure time. In water, this is created by very short (capillary) water waves, which refract a wide range of angles into the camera pixel, thus smearing the content of the pixel. The second phenomenon occurs

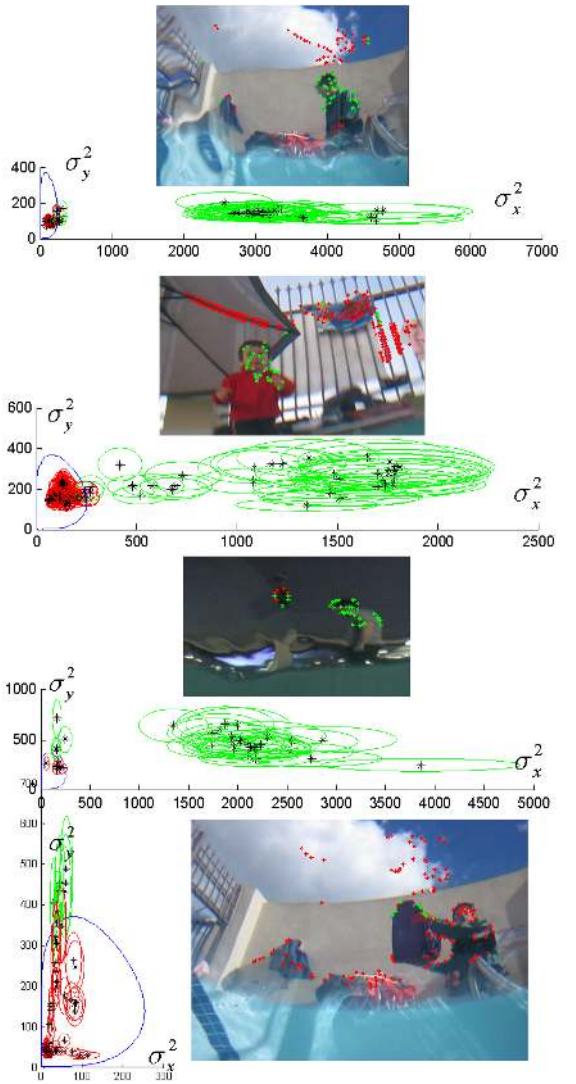


Fig. 12. The fixed empirical classifier contour in feature space, and scatter plots of several test-sets. [Upper] Horizontally walking subject. Based on the chosen confidence interval, several points were classified as dynamic instead of static. [Lower] Vertically moving object.

when viewing an airborne scene near grazing angles, while the camera is submerged. In this case, the water waves occasionally lead the water surface past the critical angle, yielding TIR. In other words, instead of acquiring the airborne object, the camera briefly captures a reflection of the underwater scene. The loss of sight of the object resembles occlusion, but here the apparent occlusion disappears as fast as it appears, or moves non rigidly.

Sudden disappearance and then re-appearance of the object makes it difficult to employ parametric optical flow [2] methods of image rectification. The natural scene complications call for representation of motion and trajectory, which is more mid-level than optical flow. We thus opt for feature tracking, since tracking has been developed in the literature to sustain geometric and radiometric distortions, as well as occlusions. We have no particular preference to any tracker. We tried several trackers. The Kernel-based

tracking [11] had been developed for non rigid objects. First, a target area (centered around an interest point) is masked with an isotropic kernel. In that area, a color histogram, which is resistant to deformations, forms a feature vector. Then, a spatially-smooth similarity function is defined. In consecutive frames, target tracking becomes a search in the basin of attraction of this function. The search is performed using the mean shift procedure [10]. The available code [16] runs in video rate, per tracked feature.

Another tracker that we tried is a large displacement optical flow (LODF) [6], [7]. In this approach, correspondences from descriptor matching are integrated into a variational optical flow model. Both trackers performed well in our experiments and provided trajectories that led to successful classification results. However, we found that LODF tracker performs somewhat better in more challenging situations.

The last tracker that we examined is the tracking-learning-detection (TLD) [23] approach. In addition to object tracking, there is a parallel learning phase. Its purpose is a detector which enables re-initialization of the tracker after it fails. The tracking is based on the Lucas-Kanade method [29]. Due to its learning and detection parts, the TLD tracking approach performs well for long-term tracking. In our short tracking period, it did not provide good results.

Fast distortions may cause the tracker to fluctuate. This may bias the estimated location variance which is our classification feature. However, these fluctuations occur for both static and dynamic objects, thus increasing the variances similarly for both. Hence, the static and dynamic feature-space clusters remain separated. We found the trackers to be robust to the aforementioned disturbances in our practical experiments, where the resulting trajectories led to highly successful classification results. However, there are extreme situations in which a tracker fails and as a result, a classification error can occur.

In very harsh fluctuations of the medium, an instantaneous video frame may become very blurred due to very fast medium motion. This motion blur may sometimes cause the tracker to pick up on a different target as a result of smearing between nearby objects. Fig. 13(a) demonstrates a situation where the tracker picks up on a person's head (object A) instead of tracking a camera base (object B). This is a result of motion smear between the two objects in frame 14. Fig. 13(b) shows the classification results. Because of the tracking error, static object B, is wrongly classified as dynamic. Perhaps this problem can be overcome by excluding extremely motion blurred frames from tracking, for example by blur detection [28].

9 Discussion

To the best of our knowledge, this work is the first to raise and characterize a problem in nature, of detecting moving objects in highly random refractive distortions. The work models the statistics of motion under dynamic distortions. We evaluate both the statistics and its error theoretically, numerically, and empirically. Features corresponding to the class of static objects were characterized, aided by the physics-based analysis of distortion statistics. A conclusion is that simple location covariance is a powerful classification feature.

The class of static objects can theoretically be characterized, if the wind/wave conditions are roughly known. Otherwise, a wide range of wind conditions can be assumed, as we did, to extend the potential class of static objects. Alternatively, the class can also be characterized by training on known static points in various conditions. In any case, tracked points whose feature vector falls outside the thresholded distribution of static objects, can be classified as being on a moving object. The rate of false negatives and positives depends on the threshold, as in any classification problem.

We believe that our approach can also be useful in milder distortions, as occurs when looking downwards into water, and in atmospheric turbulence. The approach is as follows: first use the trajectory covariance as a feature vector: the property of independent motion components and increase of variance holds in these cases as well. Then, characterize the distribution of static objects, either by physics-based calculations or by training. Then, a simple classifier is at hand.

A classification failure is likely to occur for very slow moving objects, aliasing as static in the temporal window. In nature, this situation resembles predators sneaking on prey. Feature-vectors of such slowly moving objects should be close to the class of actual static objects, leading to detection error. Objects that exhibit small movements (e.g. small hand waving) have their feature-vectors close to those of static objects.

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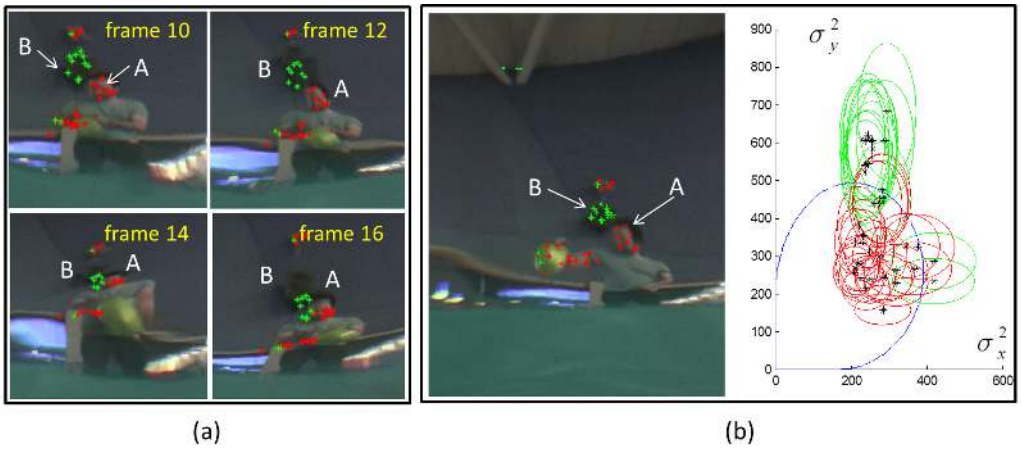


Fig. 13. Tracking failure due to motion blur. (a) Video frames showing the tracker gradually locks on a person's head (object A) instead of staying on the camera base (object B). (b) Failure classification results. Object B, which is a static object, is falsely classified as dynamic as a result of a tracker error.

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